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Determining the value of cooling in photovoltaics for enhanced energy yield

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ABSTRACT

Methods for cooling photovoltaic (PV) modules to increase their output have been proposed several times in the literature. Most of these reports describe the increase in power output achieved, but they rarely comment on the economic cost-benefit proposition. Where the economics have been considered, this has been based on measurements for the authors' specific PV system at a specific site. This means the economics are not easily interpreted for other systems at other sites.

We derive a theoretical formulation for quantifying the economic value of artificial cooling of PV modules. The formulation is not specific to any particular method of cooling. It takes as input the rate of heat removal that a cooling method can provide (in Wm^{-2} or $Wm^{-2}K^{-1}$) and determines the economic value of this cooling rate, based on variables including local solar conditions, capital cost of the system, system ventilation, plus the temperature coefficient and efficiency of the modules.

We find that the economic value of cooling PV depends strongly on the system design and local conditions, with favourable circumstances leading to a viable cost of potentially over $\$40/m^2$, however unfavourable circumstances are many times less attractive at less than $\$1/m^2$. The equations presented can be used to optimise the design of a cooling feature that is applied to a PV module or system, provided the above parameters of the cooling feature and PV system are established.

1. Introduction

An opportunity for improvement in photovoltaics that has been largely untapped, at least commercially to date, is to reduce the operating temperature of PV panels in the field. Most PV technology types, and certainly the dominant type made from crystalline silicon, experience a linear reduction in power output as the junction temperature increases. This is the result of two compounding effects; thermal broadening of the electron energy distribution, and a lowering of the effective bandgap due to thermal expansion of the atomic lattice. These effects fundamentally reduce the ability of the PV device to produce voltage, and hence are parasitic to the power output. The reduction in power output is approximately linear with temperature and hence is most commonly described in terms of a relative temperature coefficient, γ, for the maximum output power, P_{max}. The value of γ for crystalline silicon PV is typically in the range 0.40–0.45 % for every °C change above or below 25 °C.

A cooler-running PV module or system would have marketable economic value, due to an increase in power and energy output for a given nominal capacity, hence is able to attract a higher price. In this paper we present a method for determining the value of that increase in output, hence the maximum cost that should be incurred in manufacturing and incorporating the technology to produce that cooling.

Artificial cooling for photovoltaics has been studied extensively in relation to (i) concentrating PV systems, for which cooling is essential for operation, see reviews by Bahaidarah et al. (2016) and Royne et al. (2005); and also in relation to (ii) photovoltaic/thermal (PV/T) designs, where the economics include the energy collected from the heat removal system, see review by Chow (2010). In recent years a number of articles have proposed methods to cool conventional (non-concentrating) PV for the sole purpose of improving power output. Examples include radiative cooling (Habiballahi et al., 2016; Safi and Munday, 2015), spray-cooling systems (Moharram et al., 2013; Nizetic et al., 2016; Zsiboracs et al., 2016), evaporative cooling (Alami, 2014), forced air flow (Rahimi et al., 2014; Teo et al., 2012), heat pipes (Zhang et al., 2016), "sprinkling" (Bai et al., 2016) and "natural vaporization" (Ebrahimi et al., 2015). Some of these articles, particularly (Bai et al., 2016) have included some economic evaluation of the proposed

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method. To date however, all published economic analyses have been specific to the cooling design being presented, and also to the site at which it has been tested. No published work to date has described a general method to determine the economic potential of applying a certain rate of cooling to a PV module deployed at an arbitrary site.

It may at first seem trivial to determine the economic benefit of a device that provides cooling for a PV system. It is indeed true that a fixed reduction in operating temperature will result in a constant and easily calculable improvement in output, however it is by no means easy to predict what that temperature reduction will be at a given time, for an arbitrary PV device in an arbitrary installation and environment. The key parameter linking the specific and general cases is the *cooling capacity*, which describes the ability of the cooling device to remove heat per unit area, measured in Wm^{-2}, or $Wm^{-2}K^{-1}$ where appropriate. The cooling capacity is a property of the cooling device that can be refined at the design stage and then mathematically linked to its effectiveness in the field. Here we work through such a calculation to determine the cost requirements for any cooling device or technology that might be designed for non-concentrating PV power applications.

The calculation framework we propose is independent of the method used to achieve the cooling. Future technologies for PV cooling might include for example, special photonic layers for infrared reflection or enhanced radiative cooling, fixtures that increase surface area for improved convective cooling, plasmonic absorbers that capture the energy in hot carriers before it thermalises into heat, or methods that we cannot yet even conceive. Our calculation only requires that the heat removal capacity of the cooling method is known, or is calculable. That information is then used to determine the cost requirement for that method of cooling to provide a net economic benefit.

We start with the idealised “refrigeration case”, so named because it is akin to attaching a perfect refrigeration element to the rear of each solar module. For this scenario (albeit unlikely to be economically feasible), any cooling capacity can be provided, regardless of the local environment. We later extend the analysis to include two more realistic scenarios, where the cooling is proportional to the temperature difference between (i) the cell and ambient – the “ambient limited case”, and (ii) the cell and an arbitrary temperature – the “reservoir limited case”. These two cases represent more realistic types of cooling that might be proposed.

2. Methodology

In this section we build our calculation using, as a starting point, the well-known linear relationship between the PV module temperature and the prevailing solar irradiance. To aid the reader’s interpretation, the key steps are summarised as follows:

- Measure the relationship between the irradiance and the module temperature (actually, the difference between module and ambient temperatures) for PV modules of different types in the field;
- Modify this relationship in recognition that not all of the incident irradiance is absorbed by the module as heat (some heat is reflected by the module), hence introduce the term *solar heat loading*;
- Recognise that at thermal equilibrium (which applies to the PV modules in the field for most of the time), any artificial cooling applied to a module has the same effect as an equivalent reduction in the solar heat loading;
- Apply the above to determine a relationship between the rate of artificial cooling and the decrease in module temperature;
- Hence determine a relationship between the artificial cooling rate and the economic value of the energy gained – note that this depends on the solar resource and climate at the point of use of the system;
- Determine the “break-even cost” at which it becomes economically sensible to apply a given rate of artificial cooling to a given system in a given location;
- Explore the dependence of the break-even cost on the relevant input parameters.

2.1. Experimental

In order to confirm the relationship between the temperature of a PV module and the prevailing environmental conditions, measurements were made using the outdoor research facility for photovoltaics (PVORF) at the CSIRO Energy Centre in Newcastle, Australia. Specific information on that facility and this particular measurement is given in the Supplementary Information.

2.2. Determining native cooling from experimental data

In order to calculate the influence of a cooling device or technology in an arbitrary location it is necessary to identify a model that can predict the temperature of a PV module in the field. A number of such models are available, ranging from simple empirical expressions (Faiman, 2008; Ross, 1976) to rigorous thermodynamic treatments that individually account for the various heat flows that occur (Fuentes, 1987; Jones and Underwood, 2001). An excellent summary of the pertinent correlations is given in Skoplaki and Palyvos (Skoplaki and Palyvos, 2009). For the cost calculations presented here we find that a simple model, with parameters determined from careful measurements, is easily adequate. The model is based on the Ross relationship (Ross, 1976) which links the difference between module temperature, T_{mod}, and ambient temperature, T_{amb}, to the plane-of-array irradiance, G_{POA} as shown in (1). The value of the Ross coefficient k carries information about the ability of the PV module to cool naturally in its installed configuration. The typical range of k values for different PV installation types is discussed in Section 3.2.

$$T_{mod} = T_{amb} + kG_{POA} \tag{1}$$

The reader may observe that k is akin to an effective thermal resistance for the system, however that analogy is complicated by the fact that not all of the incident irradiance produces heat in the module. In order to make the model more physically meaningful, we consider only the fraction of the incident irradiance that contributes to heating the PV module, G_{heat}, and we call this the *solar heat loading*. The solar heat loading can be determined by factoring out (removing) the portion of the irradiance that is either reflected, transmitted or used in the performance of electrical work (Silva et al., 2010). The relationship between the solar heat loading and G_{POA} is therefore given in (2), where R is the AM1.5 reflectance of the PV module (transmittance is assumed to be zero) and η is the module power conversion efficiency.

$$G_{heat} = G_{POA}(1-R-\eta) \tag{2}$$

Working with the solar heat loading, rather then the irradiance, allows us to directly relate any applied cooling power to a change in the module’s observed temperature. Since for most of the time the PV module is in approximate thermal equilibrium with its environment, the relationship between G_{heat} and T_{mod} can now be used to account for any additional artificial cooling, by simply subtracting the artificial cooling power from G_{heat} and determining the new (lower) value for T_{mod}. The use of G_{heat} instead of G_{POA} means the k parameter is no longer exactly the Ross coefficient, hence from this point we will use the term k_{th} instead, see (3).

$$T_{mod} = T_{amb} + k_{th}G_{heat} - T_0 \tag{3}$$

The model in (3) contains one further modification over the previously published form in (1), with the inclusion of an offset parameter, T_0. The value of T_0 does not dominate the final result but is a potentially important correction to include, since it leads to a more consistent value for k_{th}. Including T_0 results in a more accurate linear fit to the relationship between irradiance and module temperature derived from measured data. Although neither the Ross model (Ross, 1976) nor the

similar Faiman model (Faiman, 2008) include an offset, we and others regularly observe a small negative temperature intercept in outdoor measurements of irradiance vs module temperature. We agree with other authors, e.g.(Koehl et al., 2011), that the effect is largely due to radiative cooling of the modules, and is perhaps also due to small offsets between the sensors used for the module and ambient temperatures. Although it is not strictly correct (physical) to account for radiative cooling with a simple temperature offset, the inclusion of an offset makes our measurement of the slope more accurate. The model described in (3) is now predominantly independent of the type of PV module, since it is no longer subject to differences in the heat lost in the performance of electrical work, the irradiance lost to reflection and (approximately) differences in radiative cooling due to differences in emissivity.

To determine the parameters k_{th} and T_0 in (3) we have taken measured values of T_{mod}, T_{amb} and G_{POA} from two years of outdoor data recorded using our PV Outdoor Research Facility. We extract values recorded every 10 min in conjunction with I-V curves for the module being studied; each I-V curve provides a value of P_{max} from which we can determine a value for η. A value of R for each of the two modules studied is determined using laboratory measurements of reflectance, see Supplementary Information. The R values for the polycrystalline silicon module and the CIGS module are found to be 0.059 and 0.076 respectively.

Fig. 1 shows the measured temperature delta for the polycrystalline silicon module, ΔT $(=T_{mod}-T_{amb})$ plotted against the solar heat loading, G_{heat}. Although significant scatter is present in the data, a clear linear trend is observable with a small negative offset, as expected. For this module type we obtain an average value for k_{th} of 0.0270 $^{\circ}CW^{-1}m^{-2}$ with a standard deviation of 0.0019 $^{\circ}CW^{-1}m^{-2}$. We determine k_{th} for the CIGS module (results not shown) to also be 0.0270 $^{\circ}CW^{-1}m^{-2}$ with a standard deviation of 0.0012 $^{\circ}CW^{-1}m^{-2}$. The consistency of the k_{th} value across the two different modules suggests we have effectively removed any influences of the specific module design on this parameter.

2.3. Predicting the effect of a cooling device

With k_{th} and T_0 known, (3) can now be used to determine the temperature of a PV module in an arbitrary location, provided the time-variant irradiance and ambient temperature are known for that location. We have obtained this information for different locations using climate data taken from the excellent resource available on the US Dept. of Energy website (U.S. Department of Energy, 2014). From these files we extract hourly values over the typical year for T_{amb}, diffuse horizontal irradiance (DHI) and direct normal irradiance (DNI). An isotropic sky model (Loutzenhiser et al., 2007), see (4), is used to calculate the diffuse component of G_{POA} from the DHI data and the array tilt angle, θ_T. G_{POA} is then determined by adding a cosine corrected value based on the DNI data and the solar angle of incidence at the time, using a sun position algorithm based on (Blanco-muriel et al., 2001). G_{heat} is determined from the G_{POA} value using (2), along with module specific values for R and η. This last step is easy to perform accurately for the Newcastle PVORF site, where precise values for the efficiency are measured at every data point. To keep the following analysis tractable for arbitrary locations, we can simplify the calculation of G_{heat} by using a constant value for efficiency that suits the module type, e.g. for the examples we assume η = 16.5% for wafer silicon modules. Although this now ignores the very small changes in G_{heat} that occur due to the dependence of efficiency on temperature and irradiance, we estimate the impact on the final result to be approximately 1–2% of the total value, and hence not significant in the context of the other parameter dependencies that will be observed. We re-emphasise for the reader that the assumption of a fixed efficiency for this part of the calculation is only to simplify determination of G_{heat}, and the impact of this assumption on the result is negligible.

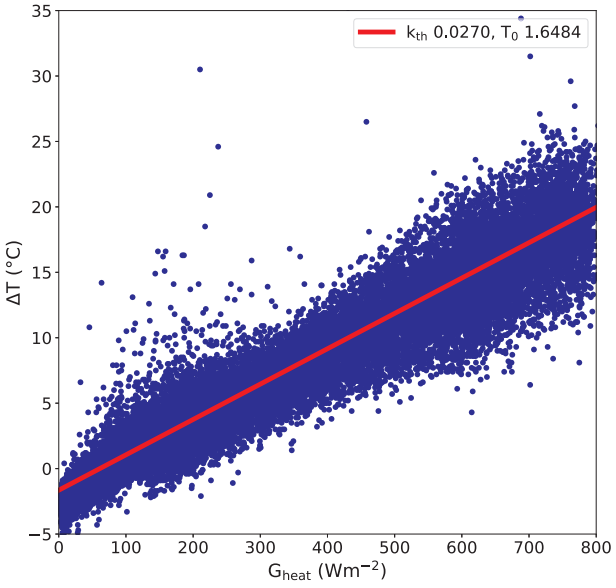


Fig. 1. The difference between the module temperature and the ambient temperature is proportional to the radiation absorbed by the module as heat.

$$G_{POA}(\text{diffuse component}) = DHI\left(\frac{1+\cos(\theta_T)}{2}\right) \tag{4}$$

To determine the new module temperature as a result of cooling, we simply modify the model to include a reduction in heating that is equivalent to the cooling we want to simulate, G_{cool}, as described in (5), where T_{mod2} is the new module temperature with cooling applied.

$$T_{mod2} = T_{amb} + k_{th}(G_{heat}-G_{cool})-T_0 \tag{5}$$

The output power at both T_{mod} and T_{mod2} (P_{max} and P_{max2}) can be calculated using (6), where T_{ref} is the reference temperature for γ, in this case the usual value of 25 °C. The increase in power due to cooling at that time is then described using (7). When integrated over the climate dataset for a year, the result is an increase in the annual energy yield that can be attributed to the application of the cooling device or technology.

$$P_{max} = \eta G_{POA}(1+\gamma(T_{mod}-T_{ref})) \tag{6}$$

$$\Delta P_{max} = P_{max2}-P_{max} = \eta G_{POA}\gamma(T_{mod2}-T_{mod}) \tag{7}$$

In the following we explore the effect of cooling through three different scenarios that are likely to be representative of real applications. The difference between the scenarios is effectively just the range of values that G_{cool} may take in (5), and how these values are influenced by some key design factors. In the first scenario it is assumed that the additional cooling is unconstrained, i.e. that a given cooling rate (in Wm^{-2}) can be provided regardless of the prevailing ambient temperature or the temperature of any heatsink attached. This scenario is referred to as the *refrigeration case*. In the second scenario the rate of cooling is assumed to be proportional to the difference between the module temperature and the ambient temperature at any given time. This would be applicable to a system that relies on convective cooling, either naturally or forced, under ambient conditions. We refer to this as the *ambient limited case*. The third scenario is a variant on the second, whereby the limiting temperature is not the (variable) ambient temperature, but instead the fixed temperature of a reservoir that is used to provide the cooling medium. We refer to this scenario as the *reservoir limited case*.

2.3.1. Refrigeration case

However impractical from a cost perspective, the case of a

refrigeration based cooling system is the simplest conceptually and hence is instructive. Here a constant amount of cooling power (per unit area) is available independently of the temperature of the module or its environment. Depending on the cooling power, this may result in the module at times being significantly below the ambient temperature. The value of G_{cool} remains constant throughout the simulated year, hence we seek to calculate the relationship between this value and the relative increase in energy yield over the year. The decrease in module temperature when cooling is applied is given by

$$\begin{aligned} T_{mod2}-T_{mod} &= (T_{amb} + k_{th}(G_{heat}-G_{cool})-T_0)-(T_{amb} + k_{th}G_{heat}-T_0) \\ &= -k_{th}G_{cool} \end{aligned} \tag{8}$$

Combining (7) and (8) and integrating over the time series of irradiance values in the climate data, $G_{POA}(t)$, leads to a value for the increase in energy yield as follows

$$\begin{aligned} \Delta E &= \int \Delta P_{max}(t)dt = \int \eta G_{POA}(t)\gamma(-k_{th}G_{cool})dt \\ &= \eta\gamma(-k_{th}G_{cool}) \int G_{POA}(t)dt \end{aligned} \tag{9}$$

The increase in energy produced is therefore linearly proportional to the fixed cooling power. Note that the result is also dependent on the location, on the basis of the integrated irradiance over the year. To determine the relative increase in energy yield compared to a base case without cooling, we use the energy generated without cooling

$$E = \int P_{max}(t)dt = \int \eta G_{POA}(t)(1 + \gamma(T_{mod}-T_{ref}))dt \tag{10}$$

and the final result is determined by taking the value of $\Delta E/E$. Since ΔE is dependent on the site insolation and E depends on both the insolation and the module temperatures, we expect to obtain slightly different results for each location studied. In Section 2.4 we use the result for $\Delta E/E$ to establish the financial benefit corresponding to a given rate of cooling. Some results for different input parameters are then presented in Section 3.

2.3.2. Ambient limited case

Perhaps a more likely scenario for a PV cooling application is one that uses a cooling medium that is limited by the ambient temperature. This would include applications that rely on evaporation, or on fluid cooling where the fluid is either air, or itself is cooled by the local atmosphere. This type of cooling device removes heat at a rate proportional to the difference between the device and ambient temperatures and therefore is best modelled using a convective heat loss term, as indicated in (11), where h is the coefficient for convective heat loss

$$G_{cool} = h(T_{mod}-T_{amb}) \tag{11}$$

Repeating the analysis in the context of a model where the system is in a thermally steady state at all times, (5) becomes

$$T_{mod2} = T_{amb} + k_{th}(G_{heat}-h(T_{mod2}-T_{amb}))-T_0 \tag{12}$$

This is an implicit equation, however it can be rearranged to determine the equilibrium solution

$$T_{mod2} = T_{amb} + \left(\frac{k_{th}G_{heat}-T_0}{1 + k_{th}h}\right) \tag{13}$$

The amount of cooling power is effectively limited by the ambient temperature. The difference in module temperature between the cooled and uncooled case thus becomes

$$\begin{aligned} T_{mod2}-T_{mod} &= \left(T_{amb} + \left(\frac{k_{th}G_{heat}-T_0}{1 + k_{th}h}\right)\right)-(T_{amb} + k_{th}G_{heat}-T_0) \\ &= \left(\frac{-k_{th}h}{1 + k_{th}h}\right)(k_{th}G_{heat}-T_0) \end{aligned} \tag{14}$$

and the change in energy yield is now

$$\Delta E = \int \Delta P_{max}(t)dt = \eta\gamma\left(\frac{-k_{th}h}{1 + k_{th}h}\right) \int G_{POA}(t)(k_{th}G_{heat}(t)-T_0)dt \tag{15}$$

$\Delta E/E$ is then determined using (15) and (10). We include this more complex scenario in the results presented in Section 3, where the value for h describes the magnitude of the applied cooling.

2.3.3. Reservoir limited case

A plausible alternative to the ambient limited case is a scenario for which the cooling is proportional to the difference in temperature between the PV module and some other value that is constant. This category would include a convective situation where a reservoir of coolant with a stable temperature is used. In this instance we begin with (16)

$$G_{cool} = h(T_{mod}-T_{res}) \tag{16}$$

where T_{res} is the temperature of a fluid in a reservoir that is taken to be large enough (or is actively controlled) such that changes in T_{res} are negligible. We therefore rewrite (5) as

$$T_{mod2} = T_{amb} + k_{th}(G_{heat}-h(T_{mod2}-T_{res}))-T_0 \tag{17}$$

This is an implicit equation, however it can be rearranged to determine the equilibrium solution

$$T_{mod2} = \frac{T_{amb} + k_{th}G_{heat} + k_{th}hT_{res}-T_0}{1 + k_{th}h} \tag{18}$$

The difference in module temperature between the cooled and uncooled case thus becomes

$$\begin{aligned} T_{mod2}-T_{mod} &= \left(\frac{T_{amb} + k_{th}G_{heat} + k_{th}hT_{res}-T_0}{1 + k_{th}h}\right)-(T_{amb} + k_{th}G_{heat}-T_0) \\ &= \left(\frac{-k_{th}h}{1 + k_{th}h}\right)(T_{amb} + k_{th}G_{abs}-T_{res}-T_0) \end{aligned} \tag{19}$$

and the change in energy yield is now

$$\begin{aligned} \Delta E &= \int \Delta P_{max}(t)dt \\ &= \eta\gamma\left(\frac{-k_{th}h}{1 + k_{th}h}\right) \int G_{POA}(t)(T_{amb}(t) + k_{th}G_{heat}(t)-T_{res}-T_0)dt \end{aligned} \tag{20}$$

$\Delta E/E$ is determined using (20) and (10). Again we include this scenario in the results in Section 3, with the magnitude of the applied cooling described by the h parameter.

2.4. Determining the economic value of increasing energy yield

There is more than one way we might place a value on an increase in energy yield for a given PV power system. We believe the most robust approach is based on the improved use of capital, rather than on the value of the electricity produced over time. This is because the latter is strongly dependent on complex factors beyond our control, such as the spot value of electricity at the time and place of use, and the long and unpredictable timeframe over which the system will operate. Our approach based on capital enhancement is simple, effective and universally applicable. Every improvement that can be made to the performance of a PV system, e.g. through the use of cooling, translates directly into a smaller number of PV modules needed to achieve the target energy yield. The financial benefit can therefore be equated to the avoided cost of those modules, plus the corresponding reductions in infrastructure and installation expenses.

To quantify the reduction in capital cost, it must be recognised that not all of the cost of a new PV system can be reduced by reducing the number of modules. For example, the inverter capacity would need to remain the same, since the cooled system will still produce the same DC power. This and other considerations for inclusion/exclusion of various cost components are described in Table 1.

Using data from recent reports on PV system prices (Fraunhofer ISE, 2016; Fu et al., 2016; Kimura and Zissler, 2016), along with our

Table 1
Cost breakdown for commercial and residential PV systems (excluding modules).

Category	Description	Included
Inverter	Cost of inverters used in system	NO[†]
Structural	Mechanical components required for installation	YES
Electrical	Electrical components required for installation	YES
Supply chain	Logistics costs associated with procurement	YES
Labour	Installation labour costs	YES
PII	Permitting, inspection and interconnection	NO[††]
Marketing	Covers cost of advertising, conducting bids etc.	NO[††]
Overheads	Covers costs associated with running the business	NO[††]
Profit	Profit margin applied by business	NO[††]

[†] The same inverter capacity will be required since the DC power produced will be the same.

[††] Remember that, at the breakeven point, we have simply replaced the cost of some of the modules with the cost of the cooling system, hence none of these costs are affected.

selections in Table 1, we establish that the fraction of non-module costs that may be reduced due to the use of fewer modules is approximately 36%, 46% and 55% for systems at residential, commercial and utility scales respectively. The same reports indicate system prices vary by country and type, and currently fall in the ranges (USD) \$1.30–3.00/W_p for residential, \$1.50–2.00/W_p for commercial and \$1.10–1.60/W_p for utility scale systems. These ranges are consistent with values reported by the US SunShot initiative (SunShot U.S. Department of Energy, 2012) and it is noted that the SunShot program has a target of US\$1/W_p by 2020. The module cost typically accounts for approximately \$0.60/W_p, hence the remainder of each range is subject to the reducible fractions indicated above. Interestingly, when we scale the non-module cost component of each range by the corresponding reducible fraction, the opposing trends in the prices (lower for larger scale) and reducible fractions (higher for larger scale) lead to a result that the absolute reducible cost is around the same for all three system size scales. The result is a price range of approximately \$0.90–1.50/W_p for the capital expenditure that could be reduced through the application of a cooling system. This parameter is included in (21) below as *PV Cost*.

A broad range of possibilities exists for a technology that could reduce the operating temperature of a PV system in the field. Examples range from a special layer added at the time of module manufacture, to a retrofit system based on, e.g. cooling fins, or a fully integrated water cooling system. Given this wide range of concepts and the need to identify a cost metric that applies to all of them, it seems most appropriate to do this on the basis of a *break-even cost* per m^2. We define the break-even cost as the cost at which it would be of equal economic value to abandon cooling and achieve the extra output by simply installing more PV modules. Since we want the break-even cost on a per m^2 basis, we must factor in the standard efficiency, η_{STC}, of the modules being used, as follows, where G_{STC} is the irradiance under standard test conditions and has the value of 1000 W m^{-2}.

$$Breakeven\ cost = \frac{\Delta E}{E} \times PVCost \times \eta_{STC} \times G_{STC} \quad (21)$$

Eq. (21) can be used to assign an economic value to a certain relative increase in PV energy yield. As an indicative example, a quick look at the data collected for the crystalline silicon modules at the Newcastle PVORF suggests that if we could cool that system to the ambient temperature all the time, we would see an energy increase of approximately 6% during the summer months and approximately 4% during the winter months. Hence, using a yearly mean value for $\Delta E/E$ of 5%, a module efficiency of 16.5% and a *PV Cost* of \$1.50, we can determine that the cooling would need to be provided at a cost of no more than \$12.37/m^2. While this may be considered a challenging cost target, we will show in the next section that more economically favourable scenarios for cooling certainly exist.

3. Results and sensitivity analysis

In Section 2 we developed expressions for the increase in energy yield for a PV system when some kind of cooling system is added, based on the geographical location (through $G_{POA}(t)$ and $T_{amb}(t)$), installation type (through k_{th} and T_0) and the temperature coefficient for output power (γ). We have also shown how the relative energy yield increase can be linked to a break-even cost in dollars (USD) per m^2 for adding the cooling system. In this section we seek to present the results across reasonable ranges for all the input parameters, to provide the reader with a sense of the overall economic value of cooling systems for PV, along with qualitative and quantitative information about the dependence on those parameters and the sensitivity to their variation. If using the results presented below to compare between cases, it should also be noted that the ambient limited and reservoir limited cases are not trivially comparable to the refrigeration case, since a slightly different metric is used to quantify the cooling performance ($Wm^{-2}K^{-1}$, compared with Wm^{-2}). This is an unavoidable consequence of the more realistic nature of those two cases.

For the examples in this section we have used $R = 0.06$, which is our measured solar reflectance value for the polysilicon wafer module, see Supplementary Information. For T_0 we have used the value that goes with our measurement of k_{th}, which is 1.64 °C. Readers looking to repeat the calculations may either provide their own measured values for k_{th} and T_0, or else simply use a literature value for k and our value for T_0, since the result is not particularly sensitive to T_0. The use of literature values for k is discussed further in 3.2.

We plot the relationship between break-even cost and applied cooling, varying one input parameter at a time while leaving the remaining parameters fixed. The values of the parameters when not being varied are: location = Newcastle, $k_{th} = 0.0270\ °CW^{-1}m^{-2}$, $\gamma = -0.5\%$, PV Cost = \$1.50, $\eta_{STC} = 16.5\%$, reservoir temperature (reservoir limited case only) = 20 °C.

3.1. Dependence on location

In Fig. 2 we present the break-even cost against a rated cooling capacity for cooling devices applied to PV modules in eleven different locations around the world. Results are clearly different for all three cases and it is emphasised that the specific values depend on the default parameters outlined above. The refrigeration case leads to a linear relationship between break-even cost and the cooling capacity as expected, however the other two cases show a levelisation of the benefit as the cooling capacity is increased. This is due to the effective limitation to cooling being imposed by the ambient temperature (or reservoir temperature). These limitations have greater impact in reducing the effectiveness of the cooling system as the cooling capacity is increased.

The astute observer will notice that the value of cooling is greater for locations known to be warmer and sunnier. This result is not as obvious as some might think, since cooling a PV module should always result in an energy yield increase, regardless of the starting temperature. In the three key result Eqs. (9), (15) and (20) it can be seen that the applied cooling contributes to a constant multiplier outside an integral of quantities that depend on the climate at the location of the PV installation. In all three cases ΔE is independent of the module temperature, however E, and therefore $\Delta E/E$, is not. The outcome is that $\Delta E/E$ trends upward with any increase in the climate parameters that influence module temperature, i.e. with the ambient temperature and the irradiance. This result is shown more clearly in Fig. 3, where the break-even cost (20 °C reservoir limited case in this example) is plotted against the mean ambient temperature and mean global horizontal irradiance for all the locations. The strong dependence on temperature indicates that hot locations lead to significantly better economics for a reservoir based PV cooling application, with a weaker dependence on the irradiance. A similar analysis performed on the ambient limited case

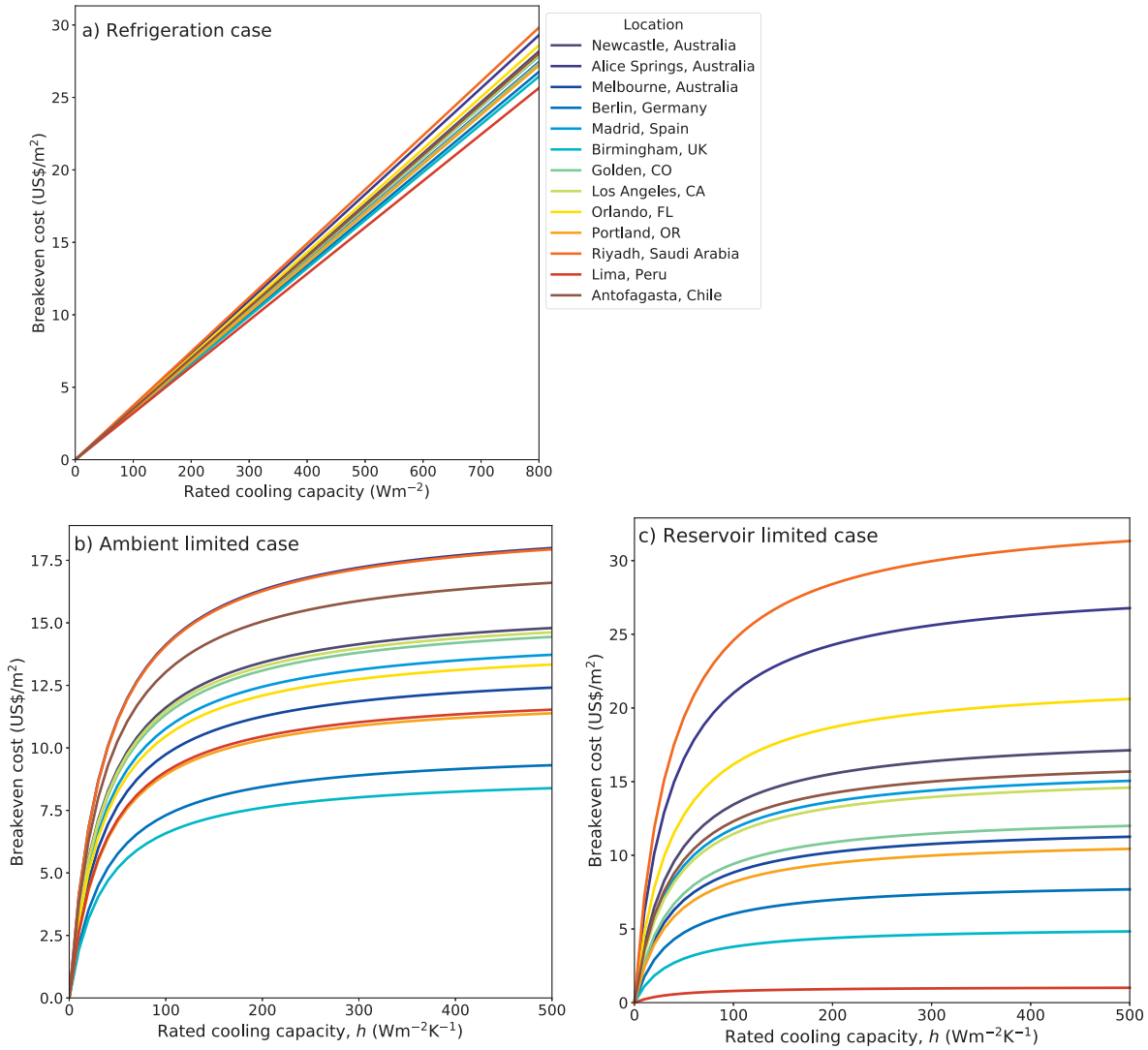


Fig. 2. Break-even cost versus cooling capacity for eleven international locations – (a) refrigeration case; (b) ambient limited case; (c) reservoir limited (20 °C) case.

shows a different relationship, where the irradiance characteristics for a particular location dictate the PV cooling economics. It is therefore important to use the approach described in Section 2.3 to correctly assess the impact of any specific cooling implementation.

3.2. Dependence on installation configuration (k_{th})

Our measured value of $k_{th} = 0.0270\,^{\circ}\mathrm{CW}^{-1}\mathrm{m}^{-2}$ for open rack, ground mounted modules is consistent with Skoplaki et al. (Skoplaki et al., 2008) which reports a value of $k = 0.0208$ for a "free-standing" system. They include a table of k values that increase as the level of natural cooling is constrained by the installation design, to a maximum reported value of 0.0563. Since we are working with k_{th}, rather than k, readers looking to apply their own k value can either treat the results below as a reasonable approximation, or else they can convert their literature value of k into k_{th} using $k_{th} = k/(1 - R - \eta)$, using the appropriate values of R and η for their nominated PV module. Reasonable values for a silicon wafer module would be $R = 0.06$ and $\eta = 0.165$. Such a conversion brings our measured value into excellent agreement with the literature value reported above. Other examples of this parameter in the literature include Green (Green, 1982), who states that a value of 0.03 is indicative of systems with an "open frame" and suggests that roof mounted systems would involve a higher value; and Durisch et al. (Durisch et al., 2007), who report Ross coefficients in the range 0.022–0.032 for five different modules on an undisclosed outdoor test facility. It is also noted that a related value is applied in the well-known PVSyst modelling package (PVsyst, 2014). Although that package uses a parameter that is the reciprocal of our k_{th}, the default values for their parameter equate to $k_{th} = 0.035$ for a free standing system and 0.067 for a "close roof mount" configuration.

In the interest of examining the economics over a representative range of k_{th} without extending to unrealistic extremes, we conclude that values between 0.02 and $0.05\,^{\circ}\mathrm{CW}^{-1}\mathrm{m}^{-2}$ are representative of most PV system designs. In Fig. 4 we plot the break-even cost against cooling capacity for k_{th} values over that range. A strong dependence on the k parameter is observed, indicating that artificial cooling is more valuable when applied to systems that are less subject to natural environmental cooling. Again we see a levelisation of the cost effectiveness of cooling for the two more realistic cases as the rated cooling capacity is increased.

3.3. Dependence on temperature coefficient

The temperature coefficient, γ, for different technology types (Emery et al., 1996) has been observed to vary in the range –0.1%/°C to –0.7%/°C, although PV modules on the market today are more likely to fall within the range –0.2%/°C to –0.5%/°C. For all three cases investigated, this parameter occurs outside the integral in the equation

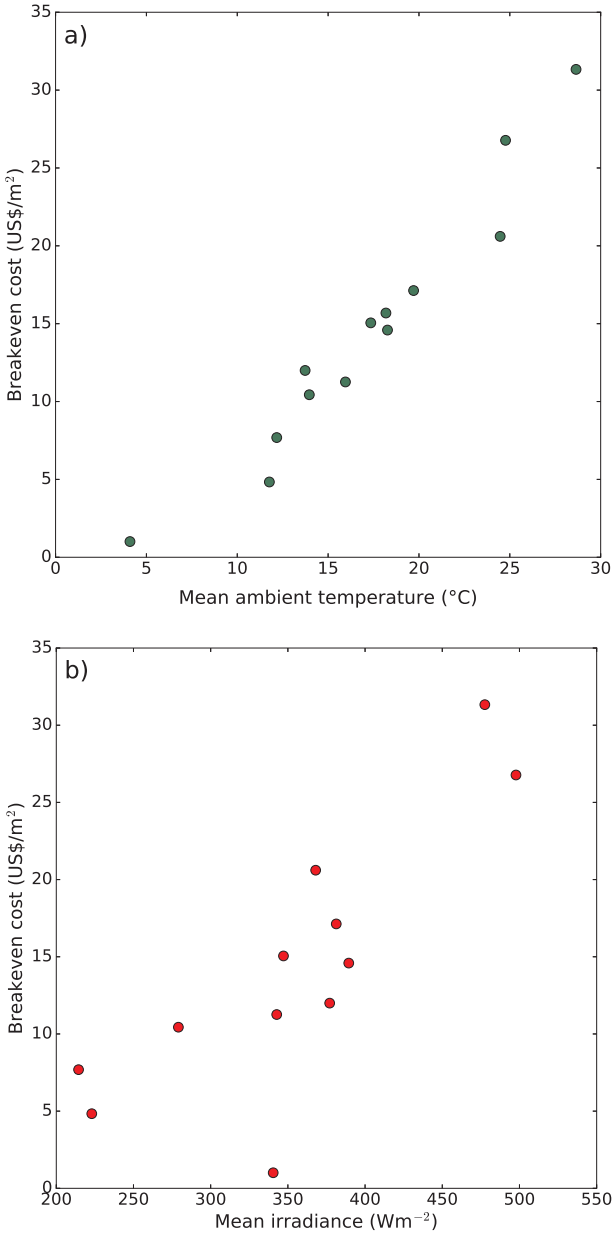


Fig. 3. Break-even cost versus (a) mean ambient temperature and (b) mean global horizontal irradiance, based on the 20 °C reservoir limited case (500 $Wm^{-2}K^{-1}$ cooling assumed) at the eleven geographic locations studied. The trend suggests much better economics for cooling in warmer, sunnier climates.

for ΔE. Although it occurs inside the integral in the equation for E, all three cases are affected in the same way. Since it is not present in (21), the result is that the break-even cost declines linearly with γ. When investigating the slope of the relationship we find that the importance of γ in determining E is small, hence the slope is approximately unity, i.e. for all practical purposes the break–even cost decreases proportionally with the temperature coefficient. In the interest of brevity we do not plot this simple relationship here.

3.4. Dependence on the cost of PV

Since we have equated the value of a cooling system to the cost of the alternative approach of increasing the system size, the break-even cost will always be directly proportional to the cost of increasing the system size, through the parameter, *PVCost*. The relationship is clearly described in (21) and hence is not plotted here.

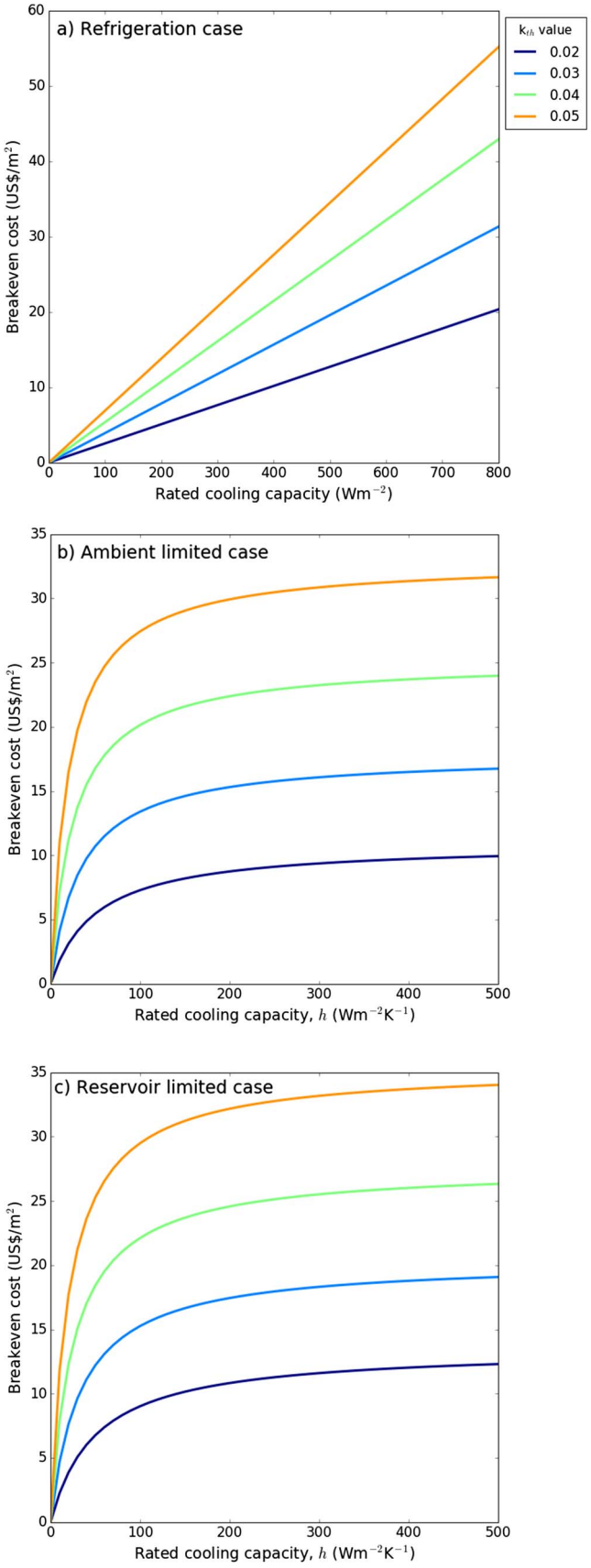


Fig. 4. Break-even cost versus cooling capacity for a range of installation configurations (values of k_{th}) – (a) refrigeration case; (b) ambient limited case; (c) reservoir limited case.

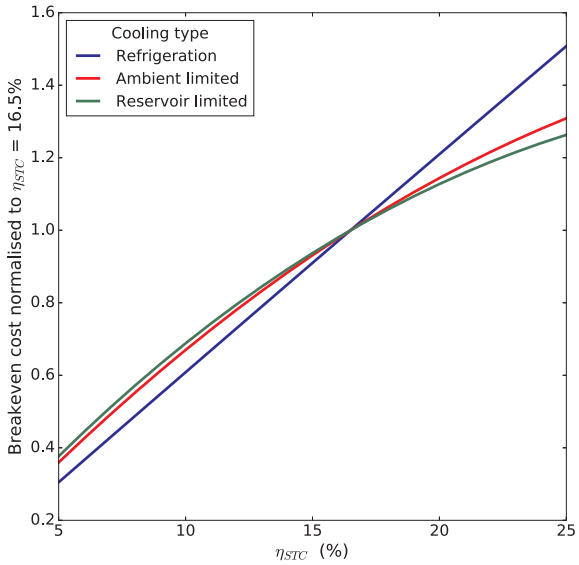


Fig. 5. Normalised break-even cost versus module efficiency for the three cases.

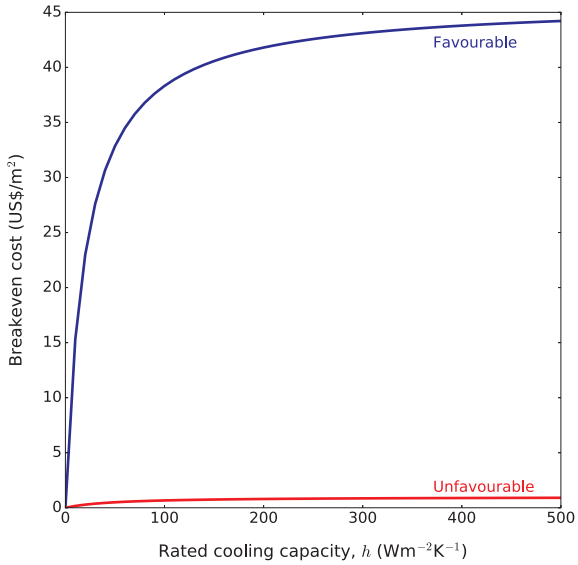


Fig. 6. Break-even cost versus cooling capacity for favourable (blue) and unfavourable (red) circumstances. The economics are better in the favourable case by one to two orders of magnitude.

3.5. Dependence on the module efficiency

The efficiency of the PV module is important, because it dictates the size of the cooling device necessary to create a given increase in power output for a given cooling capacity. This leads to a strong linear dependence of the break-even cost on the efficiency, as seen in (21). A second order dependence on efficiency is also present, through its influence on the amount of electrical work performed and the subsequent impact on the module temperature. Modules with higher efficiency make better use of the available sunlight to perform electrical work; as a result they do not heat up as much for a given set of environmental conditions. This doesn't affect the refrigeration case because in that case any additional applied cooling always changes the module operating temperature by a fixed amount. In the two limited cases however, a more efficient module will naturally operate at a temperature closer to the temperature at which the additional cooling becomes restricted, and hence there is a slight decrease of the break-even cost for increasingly efficient modules. This effect can be seen in Fig.5.

3.6. Favourable and unfavourable scenarios

In order to indicate a practical range for the absolute economic benefit of a PV cooling device, we perform the calculation using two reasonable example scenarios, one highly favourable and the other highly unfavourable. Parameters for the favourable scenario are: Location = Riyadh (Saudi Arabia), $PVCost$ = \$1.50, temperature coefficient $\gamma = -0.5\%/^{\circ}C$, $k_{th} = 0.05\,^{\circ}CW^{-1}m^{-2}$ and module efficiency $\eta_{STC} = 20\%$. Parameters for the unfavourable scenario are: Location = Birmingham (UK), $PVCost$ = \$0.90, $\gamma = -0.2\%/^{\circ}C$, $k_{th} = 0.02\,^{\circ}CW^{-1}m^{-2}$ and $\eta_{STC} = 10\%$. The break-even cost for the ambient limited cooling case is plotted against the rated cooling capacity in Fig. 6. The economics in the favourable example are quite attractive, with a break-even cost of around \$44/m^2 for cooling devices operating anywhere above 200 $Wm^{-2}K^{-1}$. In the unfavourable example however, the equivalent break-even cost is extremely low at around \$0.90/m^2. The large difference in economic viability indicates it is critical to consider the application when placing a value on a technology designed for PV cooling.

3.7. Limitations of the method

Although the method we have reported contains some simplifications for tractability, these do not reduce the accuracy of the calculation significantly. It should be noted however that our calculation presumes the provision of additional cooling is the only effect of the device or technology that is used to create it. It is reasonable to assume that in some cases, attaching a cooling device will also lead to impairment of the natural cooling that the PV module would have been subject to in its absence. For example, the incorporation of a finned heatsink onto the back of a PV module would provide the rated amount of cooling, but only after first inhibiting the rear side cooling that has been accounted for in our determination of k_{th}. Although the effect on the outcome may not be large, the reader is encouraged to bear in mind that these results refer to the gross cooling capacity of the device, as opposed to the net cooling capacity of the installed device.

Readers should also note that an accurate measurement of k for a given system requires at minimum several weeks of outdoor data, since on any given day this value may vary due to conditions such as wind. For the Newcastle (PVORF) configuration and environment we observe a good linear fit within a given day for most days, hence a reliable value of k_{th}, however the spread of daily k_{th} values across the seven month dataset has a standard deviation of approximately 20% of the mean value. The mean value over this timeframe however, shows excellent agreement with literature values as described in 3.2

4. Conclusions

We have derived equations that can be used to evaluate the cost-benefit relationship for any kind of technology designed to provide artificial cooling for a PV power system. By experimenting with the input parameters to the equations, we have found that a moderately performing cooling system must, at present, be incorporated at a cost approximately in the range US\$1/m^2 to US\$44/m^2, depending on certain parameters of the application.

The economic benefit of an artificial cooling system is found to depend strongly on the climate at the point of use, with the hottest, sunniest climates studied providing financial returns around an order of magnitude greater than for the same system in the coolest and cloudiest climates. The benefit also depends on the design of the PV system (through a recognised system configuration parameter, k) due to the influence of this on the ability of the modules to cool naturally. Better economics is associated with systems that are less able to cool naturally; a reasonable range of values for k_{th} is seen to vary the economic benefit by approximately a factor of three. Other strongly influential

parameters include the temperature coefficient of the PV output power, the module efficiency and the cost of increasing the system size.

Since the economics of cooling depend on five different parameters specific to the point of use, we encourage the reader to apply the equations to fit their own needs. The wide range of economic viability suggests that the parameters of the application should always be considered when developing a business case for adding cooling to a PV system or module.

We also observe that for realistic scenarios, the return on investment levelises significantly as cooling capacity is increased. This may be valuable information for technology developers seeking to optimise cost versus performance for a given application.

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Appendix A. Supplementary material

Supplementary data associated with this article can be found, in the online version, at http://dx.doi.org/10.1016/j.solener.2017.11.004.

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